

How Does Philosophy Become Science?
— A Formal Underlying Mechanism Based on System Ontology

Author: He Guorui*
Independent Researcher
Corresponding author: He Guorui

Email: NooEcology@outlook.com
ORCID: 0009-0006-2947-0032

Personal academic repository
(GitHub): <https://github.com/wenhe1994/wenhe001/tree/main>

Abstract

The demarcation problem in philosophy of science—how to distinguish science from non-science—has long been addressed through descriptive criteria such as verifiability, falsifiability, paradigm shifts, and research programmes. While insightful, these accounts remain at the level of historical description and lack an underlying mechanism explaining how philosophical speculation converges into scientific theory. Drawing on the axiomatic system of *System Ontology*, this paper develops a formal mechanism for the philosophy-to-science transition. We define philosophy as the *potentiality field* Φ —the set of all logically possible system laws—and science as the *truth subset* $\Sigma = \{x \in \Phi \mid \text{Valid}(x)\}$, where $\text{Valid}(x)$ is determined by four principles of legitimacy: **Logical**

Legitimacy (consistency), **Observability Legitimacy** (the existence of a principled codifiable interaction channel with the cognitive framework), **Formal Legitimacy** (structurable, unambiguous expression), and **Boundary Legitimacy** (the absence of illegitimate infinite nesting and the presence of actual upper and lower bounds in application). Through historical case studies—the transition from natural philosophy to classical mechanics, from alchemy to chemistry, and Darwinian evolution as a weakly formalized science—we demonstrate how the four principles operate in actual scientific practice. We further clarify the three levels of formalization (weak, semi, strong) and argue that mathematics is a sufficient but not necessary condition for science. Finally, we complete the self-referential closure of the meta-theory: the theory of how science becomes science is itself examined by the four principles via an **external judgment stance**, avoiding vicious circularity. This framework transforms the demarcation problem from static description into a dynamic convergence mechanism, providing a unified meta-theoretical foundation for *Noetic Ecology* and *Civilization Dynamics*.

Keywords: System Ontology; demarcation problem; philosophy-to-science convergence mechanism; four principles of legitimacy; levels of formalization; self-referential closure; meta-theory

1. Introduction

1.1 The Problem Stated

One of the central missions of philosophy of science has been to answer the question “What is science?”—the so-called demarcation problem. From the logical positivists’ principle of verifiability, to Karl Popper’s falsificationism, to Thomas Kuhn’s paradigm theory, Imre Lakatos’s methodology of scientific research programmes, and Paul Feyerabend’s epistemological anarchism, the philosophy of science has amassed a rich and profound intellectual legacy over the past century.

Yet a fundamental question remains unanswered: **How does philosophical speculation converge into scientific theory?** In other words, by what operational process does a conjecture about the fundamental nature of the world, a metaphysical framework, or an intuition emerging from personal experience become public, testable, formalizable knowledge that can be reliably reproduced across subjects?

The answers offered by traditional philosophy of science remain largely at the level of **description**. Popper tells us that scientific propositions must be falsifiable—a demarcation criterion, but one that does not explain how a proposition born of philosophical speculation undergoes the dynamic process from “being falsifiable” to “being confirmed/falsified.” Kuhn reveals that scientific activity is governed by paradigms and that paradigm shifts are revolutionary—yet this remains a phenomenological description of scientific history rather than a mechanistic account of how paradigms emerge from philosophical possibilities and become legitimized. Lakatos describes the evolution of theories in terms of “hard core” and “protective belt,” but he does not provide operational criteria for how the hard core itself acquires legitimacy. Feyerabend dissolves methodological dogmatism but slides into relativism, offering no guidance for concrete knowledge production.

This predicament—**more description than mechanism**—is no accident. Its deep cause lies in traditional philosophy of science’s persistent avoidance or inability to handle **logical self-reference**. When a theory attempts to answer “how science is possible,” it must also answer whether it itself is science and on what grounds. This self-referential structure tends to lead to paradox in formal systems (as Gödel’s incompleteness theorems reveal), which is why most philosophical traditions have treated it as a forbidden zone to be circumvented. However, if we recognize that human cognition itself is a system capable of safely using self-reference (i.e., able to “think about its own thinking” without collapsing), then a theory about “how science becomes science” should not be excluded from science—on the contrary, it should be the highest form of science, namely **meta-science**.

This is precisely the path this paper attempts to open.

Recent developments in cognitive science, complex systems theory, and artificial intelligence compel us to revisit a set of ancient yet urgent questions: Where does human creative intuition come from?

Why do paradigm revolutions always erupt at the boundaries of cognition? How does a theory “collapse” from philosophical speculation into a computable, testable science? If we rely solely on the descriptive frameworks of traditional philosophy of science, these questions can no longer be answered satisfactorily. What we need is an underlying mechanism capable of **handling self-reference, accommodating evolution, and being operationalizable**.

1.2 The Task of This Paper

The task of this paper is to construct a **formal underlying mechanism of how philosophy becomes science**, grounded in the axiomatic system of *System Ontology*. Specifically, we will accomplish the

following:

First, we redefine the relationship between philosophy and science. We conceive philosophy as the **potentiality field** Φ —the set of all logically possible system laws, guaranteed by Meta-Axiom 0 (the principle of self-organization)—and science as the **truth subset** $\Sigma = \{x \in \Phi \mid \text{Valid}(x)\}$ selected from the potentiality field, where $\text{Valid}(x)$ represents the legitimacy criteria for scientific admission. Second, from the axiomatic system of *System Ontology*, we derive **four principles of legitimacy** for scientific admission: **Logical Legitimacy** (consistency), **Observability Legitimacy** (the existence of a principled codifiable interaction channel with the cognitive framework), **Formal Legitimacy** (structurable, unambiguous expression), and **Boundary Legitimacy** (the absence of illegitimate infinite nesting and the presence of actual upper and lower bounds in application). These four principles together constitute the **convergence mechanism** from philosophy to science. Third, we test this mechanism through historical case studies. We select three representative cases—the transition from natural philosophy to classical mechanics, the transition from alchemy to chemistry, and Darwinian evolution as a weakly formalized science—to demonstrate how the four principles operate in actual scientific practice. Fourth, we clarify the **three levels of formalization** (weak, semi, strong) and their relationship to mathematics, showing that “formalization” is not equivalent to “mathematization,” and locate the present theory within this spectrum. Fifth, we complete the **self-referential closure** of the meta-theory: the theory of how science becomes science must itself be examined by the four principles and, by adopting an **external judgment stance**, avoid vicious circularity. This is not merely self-justification but the ultimate proof of its own legitimacy.

1.3 Structure of the Paper

This paper is organized into eight chapters. Chapter 2 systematically reviews the historical legacy of the demarcation problem in philosophy of science, identifies its descriptive limitations, and articulates the incremental contributions of the present theory. Chapter 3 constructs the theoretical framework, providing set-theoretic definitions of philosophy and science and deriving the four principles of legitimacy from the axioms of *System Ontology*. Chapter 4 tests the mechanism through three historical case studies. Chapter 5 discusses the three levels of formalization and the role of mathematics, clarifying the position of the present theory. Chapter 6 completes the self-referential closure of the meta-theory. Chapter 7 engages systematically with the traditions of Popper, Kuhn, Lakatos, Feyerabend, and Gödel, demonstrating how the present theory integrates and transcends them. Chapter 8 concludes with a summary of contributions and directions for future research.

The argument of this paper follows a strictly deductive path from axioms to theorems, from theory to cases, and from mechanism to self-reference. Our aim is to show that the boundary between philosophy and science is not an impassable chasm but a convergent channel that can be formally described and operationally navigated through the **four principles of legitimacy**. And the very unveiling of this channel itself constitutes a self-referential closure, making the meta-theory an integral part of the very scientific evolution it describes.

2. Literature Review: The Legacy of the Demarcation Problem and the Incremental Contribution of the Present Theory

Over the past century, the philosophy of science has accumulated a rich intellectual heritage around the demarcation problem—the question of what distinguishes science from non-science. From the logical positivists’ principle of verifiability, to Popper’s falsificationism, to Kuhn’s paradigm theory, Lakatos’s methodology of scientific research programmes, and Feyerabend’s epistemological anarchism, each stage has illuminated different facets of science, yet each has also revealed unavoidable limitations. This section systematically traces this intellectual lineage and articulates how the **four-principle convergence mechanism** proposed in this paper achieves a transition from *description* to *mechanism*.

2.1 Logical Positivism: The Principle of Verifiability and Its Limitations

Logical positivism (represented by the Vienna Circle) sought to demarcate science from metaphysics through the **principle of verifiability**: a proposition has cognitive meaning only if it can in principle be empirically verified. This principle proved powerful in excluding traditional metaphysics (such as “absolute spirit” or “thing-in-itself”), but it soon ran into difficulties.

First, the principle of verifiability could not handle universal statements. Scientific laws often take the form “all swans are white,” and such statements cannot be fully verified by finite empirical observations. Second, it placed theoretical entities (such as electrons, genes) in an awkward position: they can neither be directly verified nor excluded from science. Third, the principle of verifiability itself cannot be empirically verified, leading to a self-referential predicament.

Logical positivism declined in the mid-twentieth century, but its central concern—a clear, operational criterion for scientific admission—remains the origin of the demarcation problem. The **observability legitimacy** principle introduced in this paper (see Section 3.2) inherits logical positivism’s emphasis on empirical grounding, but replaces “verification” with “the existence of a principled codifiable interaction channel with the cognitive framework,” thereby avoiding the difficulties posed by universal statements and theoretical entities.

2.2 Popper: Falsificationism

In *The Logic of Scientific Discovery* (1934), Popper proposed **falsificationism**: the hallmark of science is not verifiability but falsifiability. A theory is scientific only if it rules out certain possible observations. Falsificationism solved the problem of universal statements—a proposition like “all swans are white” can be falsified by finding a single black swan—and placed theories in a testable situation rather than seeking unattainable verification.

Falsificationism has had a profound influence on scientific methodology, yet its limitations are equally evident. First, many important theories in the history of science were not directly falsifiable in their early stages (e.g., Darwinian evolution initially lacked direct observational evidence), yet they were recognized as scientific. Second, falsificationism cannot explain the resilience of theories in the face of anomalous evidence in actual scientific practice: a theory is seldom abandoned after a single counterexample; scientists adjust auxiliary hypotheses or refine measurement methods rather than immediately discarding the core theory. Third, falsificationism remains a demarcation criterion rather than a dynamic account of how philosophical possibilities converge into science.

The **observability legitimacy** principle in this paper absorbs the core insight of falsificationism—scientific propositions must maintain a testable connection with the empirical world—but extends it to a more general condition: a scientific proposition need only have a principled codifiable interaction channel with the cognitive framework, rather than being required to be immediately falsifiable. This

extension accommodates the developmental stages of theories and leaves room for the discussion of levels of formalization (see Chapter 5).

2.3 Kuhn: Paradigm Theory

In *The Structure of Scientific Revolutions* (1962), Kuhn fundamentally transformed the discussion of the demarcation problem. He argued that science is not governed by a set of eternal rational rules, as logical positivism or falsificationism would have it, but evolves through alternations between **normal science** and **scientific revolutions**. Normal science is governed by a **paradigm**—a set of theoretical frameworks, fundamental assumptions, research methods, and exemplary achievements shared by a scientific community. During normal science, scientists work on “puzzle-solving” within the paradigm without questioning the paradigm itself. When the paradigm can no longer accommodate accumulated anomalies, a crisis ensues, culminating in a scientific revolution in which a new paradigm replaces the old.

Kuhn’s contribution lay in revealing the social structure and historical dynamics of science, but he did not provide an operational mechanism to explain how a new paradigm is selected from philosophical possibilities. His account remains descriptive and historically inductive, rather than deductive and operationalizable. Moreover, Kuhn’s account of paradigm change carries a relativistic undertone—paradigms are incommensurable, and the standard of scientific progress depends on community consensus rather than transcendent rational criteria.

This paper responds to Kuhn in two steps. First, we subsume “paradigm” under the category of **cognitive framework** (see *System Ontology*, Axiom 5), and identify paradigm change as a macroscopic manifestation of **cognitive escape** (see *Noetic Ecology*, Theorem D23). Second, the four principles proposed here provide operational criteria for the legitimacy of a paradigm: to be scientific, a paradigm must satisfy logical, observability, formal, and boundary legitimacy. This framework explains why a new paradigm replaces an old one (because it performs better on the four principles) and avoids relativism (because the four principles are independent of any particular community’s consensus).

2.4 Lakatos: Methodology of Scientific Research Programmes

In *The Methodology of Scientific Research Programmes* (1970), Lakatos attempted to reconcile Popper and Kuhn. He conceived the development of theories as the evolution of **scientific research programmes**: each programme consists of an inviolable **hard core** and an adjustable **protective belt**. Scientists adjust the protective belt to cope with anomalies rather than immediately abandoning the hard core. A programme is “progressive” or “degenerating” depending on whether it continues to predict novel facts.

Lakatos’s work advanced the methodological understanding of scientific evolution, yet he still did not provide an underlying mechanism for how the hard core itself acquires legitimacy. His account remains at the level of *evaluating* theories rather than explaining how a theory is born from philosophical speculation and legitimized. Moreover, his judgments of “progressive” versus “degenerating” rely on retrospective historical evaluation, lacking predictive power.

In the present framework, a scientific research programme is assimilated into the more fundamental concept of a cognitive framework: the hard core corresponds to the axiomatic layer (the core presuppositions of the framework), while the protective belt corresponds to auxiliary hypotheses and formal expressions. The evolution of a programme is the process by which a cognitive framework, driven by structural tension (see *System Ontology*, Theorem 4), undergoes screening by the four

principles and elevates its level of formalization. This perspective transforms Lakatos's descriptive model into an operationalizable dynamic mechanism.

2.5 Feyerabend: Epistemological Anarchism

In *Against Method* (1975), Feyerabend advanced a radical **epistemological anarchism**: there is no single scientific method; “anything goes.” He criticized all forms of methodological dogmatism, arguing that science should accommodate every possible mode of inquiry, and any rule may be broken under certain conditions.

Feyerabend's work deconstructed entrenched methodological myths in philosophy of science, but its relativistic stance left the theory unable to guide practice. If “anything goes,” then no distinction can be made between science and pseudoscience, depriving science education, research funding, and public policy of a rational basis.

This paper resonates deeply with Feyerabend: we do not believe there is a single, eternal scientific method. However, we also argue that scientific activity is not entirely unconstrained—any knowledge recognized as science must satisfy, in terms of its *conditions of existence*, the four principles of legitimacy. These principles are not “methods” but **thresholds**: they do not prescribe how science should be produced, only the minimal conditions that scientific knowledge must satisfy. Diverse modes of inquiry are acceptable as long as the knowledge they produce passes the four principles. This stance respects methodological pluralism while providing a clear boundary between science and non-science, avoiding relativism.

2.6 The Incremental Contribution of the Present Theory

In summary, the evolution of the demarcation problem in philosophy of science reveals a deepening of thought—from logical rules to historical description, from single criteria to pluralistic frameworks. Yet this lineage has remained at the level of *description*, failing to uncover the underlying mechanism by which philosophical speculation converges into scientific theory.

The theoretical framework proposed in this paper offers the following incremental contributions:

First, from description to mechanism.

Traditional theories (Popper, Kuhn, Lakatos) tell us *what* science is like (falsifiable, paradigm-governed, with hard core and protective belt). This paper tells us *how* science is selected from philosophical possibilities: through the four-principle legitimacy screen, converging the potentiality field Φ into the truth subset Σ .

Second, from demarcation to convergence.

The demarcation problem presupposes a static opposition between science and non-science. This paper adopts a dynamic perspective: philosophy is the unfolding of possibility, science is the convergence of possibility. There is no impassable chasm between them, only a process of selection via the four principles.

Third, from external rules to internal conditions.

Traditional theories treat the criteria of science as externally imposed rules (e.g., verifiability, falsifiability). The four principles in this paper are derived directly from the axiomatic system of *System Ontology*; they are intrinsic conditions that any stable system must satisfy. Hence, science is not a matter of conforming to external norms but of meeting the requirements of existence itself.

Fourth, from avoidance of self-reference to accommodation of self-reference.

Traditional philosophy of science, fearing the paradoxes of self-reference, has largely avoided meta-

theoretical questions. This paper confronts self-reference head-on and completes the self-referential closure of the meta-theory: the theory of how science becomes science is itself examined by the four principles and, by adopting an external judgment stance, escapes vicious circularity. This not only makes the present theory a part of science but elevates it to **meta-science**—the science of how science becomes science.

Fifth, providing a meta-theoretical foundation for applied theories.

The mechanism established in this paper serves as the meta-theoretical premise for *Noetic Ecology*'s “cognitive expansion theorem” and *Civilization Dynamics*' “phase transition model.” It provides a formalized underlying logic for understanding how individual cognition breaks through frameworks and how civilizational paradigms undergo holistic transitions.

In the following chapters, we will derive this mechanism from the axioms of *System Ontology* and test its validity through historical case studies.

3. Theoretical Framework: From Axioms to Convergence Mechanism

In the preceding chapters, we surveyed the historical legacy of the demarcation problem in philosophy of science and identified its fundamental limitation: more description than mechanism. The task of this chapter is to construct a **formal underlying mechanism of how philosophy becomes science**, grounded in the axiomatic system of *System Ontology*. We will first redefine the relationship between philosophy and science, then derive the four principles of legitimacy from the axioms, and finally articulate the independence and completeness of the four principles.

3.1 Philosophy = Potentiality Field Φ , Science = Truth Subset Σ

3.1.1 Philosophy: The Potentiality Field Φ

Meta-Axiom 0 (The Principle of Self-Organization) states: Being itself contains an eternal, non-teleological, far-from-thermodynamic-equilibrium fundamental tendency. This tendency constantly attempts to instantiate various possible system structures from an infinite field of possibilities. This “infinite field of possibilities” is the **potentiality field**, denoted Φ .

The potentiality field contains all logically possible system laws—different combinations of physical constants, different mathematical structures, different cognitive frameworks, different forms of social organization. It does not presuppose whether these possibilities can be instantiated in reality, nor does it require them to be empirically tested. **Philosophy** is precisely the exploration of this potentiality field. The task of philosophy is to **unfold the space of possibilities**, to propose new concepts, new categories, new frameworks of inquiry. In this sense, philosophy precedes science—it supplies the candidates from which science selects.

3.1.2 Science: The Truth Subset Σ

Not everything that is logically possible qualifies as “science.” Science imposes a stricter admission standard: a proposition or theory must not only be logically consistent but must also acquire a certain kind of “legitimacy certification” through interaction with reality.

We denote by Σ the **truth subset** selected from the potentiality field:

$$\Sigma = \{x \in \Phi \mid \text{Valid}(x)\}$$

where $\text{Valid}(x)$ means “ x passes the legitimacy criteria for scientific admission.” **Science** is the systematic exploration of Σ . It is not an arbitrary pruning of the space of possibilities but a legitimate convergence carried out according to an internal set of constraints (the manifestation of Meta-Axiom 1 at the cognitive level).

Supplementary note: $\text{Valid}(x)$ includes not only the four principles to be elaborated below but also implicitly requires that the system corresponding to x possesses the possibility of being instantiated in reality—that is, that the product of self-organizing tendency and environmental permissibility can in principle exceed the binding force (see *System Ontology*, Theorem 5: The Critical Theorem of System Emergence). Purely “logical possibilities” that can never cross the emergence threshold remain in the residual part of the potentiality field and do not enter Σ .

3.2 Derivation of the Four Principles of Legitimacy from the Axioms

The four principles of legitimacy are not externally imposed norms but systemic conditions derived from the axiomatic system of *System Ontology*—conditions that any scientific theory must satisfy to acquire “existential legitimacy.” Each principle has its own axiomatic source, and together they constitute a complete convergence mechanism.

3.2.1 Logical Legitimacy

Statement: A scientific theory must be internally consistent and free from contradiction.

Axiomatic source: Derived from **Meta-Axiom 1 (The Principle of Systemic Constraint)**. Meta-Axiom 1 states: any stably existing system must contain within itself a set of self-constraining meta-rules that eliminate the “vicious self-reference” which would lead to logical collapse. In propositional systems, logical contradiction is precisely a form of vicious self-reference—a system containing a contradiction can derive any proposition, thereby losing all informational content and failing to exist as a stable body of knowledge.

Explanation: Logical legitimacy is the minimal condition for any system to exist. This is not because “the real world disallows contradiction” (the real world may be full of paradoxical phenomena), but because **a self-contradictory theory cannot function as an effective cognitive framework**—it cannot provide reliable guidance for action, nor can it form stable intersubjective consensus. Therefore, any claim to science must first pass the test of logical consistency.

3.2.2 Observability Legitimacy

Statement: A scientific proposition must be capable of interacting with observer systems (including measuring instruments, experimental apparatus, sensory organs) and leaving detectable effects. Purely speculative, unobservable propositions do not belong to science.

Axiomatic source: Derived from **Axiom 3 (The Principle of Existence and Interaction)**. Axiom 3 states: all objects that can be spoken of, deduced, and interacted with are “systems”; the sole proof and mode of existence of a system is its interaction with other systems; a system without interaction is equivalent to non-existence.

Explanation: “Observable” does not mean “already observed.” A proposition need only in principle be capable of leaving detectable effects (even if technically infeasible at present) to satisfy observability legitimacy. For example, black hole event horizons were confirmed through gravitational effects long before they were directly “seen.”

Distinguishing “existence” from “scientific observability”

Axiom 3 defines “existence” in the ontological sense: to exist is to interact with any system. Observability legitimacy defines “observability” in the epistemological sense of science: the existence of a principled codifiable interaction channel with the human cognitive framework (or with a designable measuring instrument).

A system may interact with other systems (and thus exist ontologically), but if it happens to have no effective interaction channel with our cognitive framework, it does not satisfy observability legitimacy and cannot become an object of current science. This distinction avoids the misunderstanding that “unobservable = non-existent.”

Observability legitimacy excludes purely metaphysical speculation. Propositions that have no causal connection with any possible experience (such as “there exists a parallel universe outside our own that cannot be perceived in any way”) are meaningless for science—they belong to the residual part of the potentiality field and can never enter Σ .

3.2.3 Formal Legitimacy

Statement: Scientific knowledge must be expressible using symbolic systems (language, mathematics, diagrams) and possess publicly operable form.

Axiomatic source: Derived from **Axiom 2 (The Principle of Material/Noetic Duality)**. Axiom 2 states: any system simultaneously manifests as both a material entity and a noetic pattern; the noetic state is the mode, rule, and representation of information organization within and between systems. The noetic state must be capable of being encoded and transmitted in some form.

Explanation: Formal legitimacy is not equivalent to mathematization. Much scientific knowledge (e.g., biological taxonomy, geological description) initially took the form of natural language and classificatory frameworks, yet possessed clear structures and rules, making it teachable, learnable, and criticizable. Mathematics is the highest-order medium of formalization, but not the only form. The core requirement of formal legitimacy is that the expression of knowledge must be unambiguous, repeatable, and operable by different cognitive carriers (different minds, different cultures, different eras). An “insight” that exists only in personal intuition, however logically consistent and observable it may be, cannot become science if it cannot be expressed symbolically.

3.2.4 Boundary Legitimacy

Statement: A scientific theory must acknowledge its own domain of applicability and cannot presuppose infinite nesting or absolute completeness. This consists of two levels:

- **Logical level requirement:** The theory must not contain logical structures of “infinite nesting” or “infinite subdivision” (e.g., “the set of all sets”).
- **Empirical level requirement:** When applied to concrete systems, the theory must specify the actual upper and lower bounds of those systems (e.g., physical laws hold only within specific energy scales).

Axiomatic source: Derived from **Axiom 5 (The Principle of Framework and Constraint)** and **Theorem 2 (The Theorem of Finite Hierarchical Nesting)**. Axiom 5 states: any system must possess a “cognitive framework” that defines itself, filters interactions, and guides internal processes; without a framework, the system dissolves into chaos. Theorem 2 states: any stably observable real system must have finite hierarchical nesting, because a system with infinite nesting cannot form a stable cognitive framework.

Explanation: Boundary legitimacy reveals that the “finitude” of scientific theories is not a defect but a

condition of their stable existence. A theory that claims to “explain everything” would precisely lose legitimacy by attempting to transcend boundaries—because it could not handle its own framework problems.

In science, boundary legitimacy manifests as:

- **Domain of validity:** Newtonian mechanics is valid at low speeds and macroscopic scales; relativity at high speeds and strong gravity; quantum mechanics at microscopic scales. No theory is absolutely universal.
- **Exclusion of infinities:** When infinities appear in scientific theories (e.g., divergences in quantum field theory), it usually indicates that the theory is breaking down at its boundaries, requiring renormalization or a new theory.
- **Acceptance of incompleteness:** As Gödel’s theorems reveal, any sufficiently complex formal system cannot prove its own completeness and consistency within itself. Science must accept this incompleteness as a driving force for evolution rather than a defect.

3.2.5 The Hierarchical Difference in the Legitimacy of Self-Reference

This conclusion can be directly derived from the **Law of Systemic Hierarchy** in *System Ontology*. The Law of Systemic Hierarchy (derived from Axiom 1 and Theorem 2) states: lower-level components combine to emerge higher-level systems; the higher level depends on the lower level, but the lower level does not depend on the higher level. Hence, lower-level formal systems prohibit self-reference to maintain stability, while higher-level cognitive systems, by objectifying the lower level, can safely employ self-reference. The elaboration follows:

In the tradition of philosophy of science, “logical self-reference” has often been regarded as the source of paradox and thus strictly avoided. Gödel’s theorems, Russell’s paradox, the halting problem—all demonstrate that in formal systems (such as mathematics, computer programs), self-reference can lead to collapse. Yet human cognitive systems safely use self-reference: we can “think about our own thinking,” simulate different possibilities in our minds, and examine our own cognitive frameworks without falling into paradox. How is this contradiction resolved?

The answer lies in **the hierarchical difference of systems**.

Lower-level formal systems: self-reference leads to collapse

Formal systems (such as mathematical axiomatic systems, programming languages) are **non-hierarchical planar structures**. All their rules apply at the same level, with no distinction between “meta-level” and “object-level.” When self-reference occurs (e.g., “this sentence is false”), the system has no safe “anchor point” for the self-reference and thus falls into a vicious cycle. This is precisely the essence revealed by Gödel’s theorems: any sufficiently powerful formal system is either incomplete or inconsistent—this is the inherent limitation of lower-level systems.

Higher-level cognitive systems: self-reference becomes legitimate through hierarchy

The human cognitive system is **hierarchically structured**. According to *System Ontology* Axiom 1 (The Dynamical Principle of Logical Self-Reference) and Theorem 2 (The Theorem of Finite Hierarchical Nesting), the cognitive system comprises at least three levels:

- **Lower level:** The biological-instinctual layer (emotions, instincts), which executes rigid rules and does not permit self-reference.
- **Middle level:** The subconscious processing layer (pattern recognition, intuition), which can perform limited self-reference but remains constrained by the lower level.
- **Upper level:** The meta-cognitive layer (reflection, philosophical inquiry, theory construction),

which can safely perform self-reference on the lower levels—objectifying the lower-level formal systems for simulation, deduction, and revision without causing systemic collapse. This hierarchical structure makes “safe self-reference” possible: self-reference at the upper level operates on **objectified lower-level systems**, not on its own level. Just as we can use natural language to talk about natural language (separating meta-language from object-language) without falling into semantic paradox, cognitive systems separate levels to transform self-reference from “vicious cycle” into “meta-cognitive tool.”

Significance for the philosophy-to-science convergence mechanism

This insight directly supports the **boundary legitimacy** principle in this paper. The logical-level requirement of boundary legitimacy—“the theory must not contain logical structures of infinite nesting or infinite subdivision” (derived from Theorem 2)—echoes the prohibition of vicious self-reference in lower-level formal systems. The meta-theory constructed in this paper—the theory of how science becomes science—can safely perform self-reference (examining itself by the four principles) precisely because it operates at the meta-level, taking object-theories (knowledge about particular sciences) as its objects rather than performing self-reference within its own level.

Therefore, the self-referential closure of the present theory (Chapter 6) is not a vicious circle but leverages the hierarchical difference of cognitive systems: the author and reader, standing at the meta-level, perform an **external judgment** on the theory as an object. This operation is logically legitimate because it does not violate the prohibition of self-reference in lower-level formal systems but confines self-reference to the safe range of the meta-level.

Connection to System Ontology

This view is formally expressed in *System Ontology* Axiom 1 and Theorem 2. Interested readers may consult that work.

3.3 Independence and Completeness of the Four Principles

3.3.1 Independence

The four principles each correspond to different axiomatic sources and cannot be reduced to one another:

- **Logical legitimacy** excludes contradiction but cannot exclude unobservable propositions (a logically consistent proposition may be completely unobservable).
- **Observability legitimacy** excludes unobservable propositions but cannot guarantee formal expression (an observable phenomenon may exist only in oral tradition).
- **Formal legitimacy** guarantees public operability but cannot ensure clear boundaries (a formally complete theory may presuppose infinite nesting).
- **Boundary legitimacy** prevents infinite nesting but cannot alone ensure logical consistency or observability.

The four principles are independent and together constitute a complete screening network for scientific admission.

3.3.2 Completeness

Within the axiomatic system of *System Ontology*, these four principles are **necessary and sufficient** for scientific admission:

- **Necessity:** A scientific theory lacking any one of these principles would fail to be a stable cognitive system. Without logical legitimacy, it would self-destruct; without observability

legitimacy, it would be disconnected from experience; without formal legitimacy, it could not be transmitted or tested; without boundary legitimacy, it would fall into the chaos of infinite nesting.

- **Sufficiency:** Any proposition proposed from the potentiality field that simultaneously satisfies the four principles possesses the qualification to become science. It may still require further empirical research to confirm its truth value, but in terms of “legitimacy,” it is no longer purely philosophical speculation.

3.4 Methodological Note: Finite Convergence

This paper deduces the four principles from the axioms of *System Ontology* and then applies them to test historical cases. This path embodies a methodological principle: **finite convergence**.

The construction of theory often faces the predicament of **infinite regress**—axioms themselves require grounding, and that grounding requires more fundamental axioms, ad infinitum. The approach taken here is to **anchor the validity of axioms in the testing of historical cases**, rather than pursuing an infinite regress of absolute self-justification.

The deduction of the four principles terminates at the meta-axioms and system axioms of *System Ontology*. The “legitimacy” of these axioms themselves is not proven by even more fundamental axioms but is secured by three considerations:

1. **Internal consistency:** The axiomatic system contains no contradictions.
2. **Explanatory power:** It can uniformly explain the convergence phenomena in the history of science (classical mechanics, chemistry, evolution).
3. **Empirical anchoring:** The historical cases serve as “factual anchors,” demonstrating that the conclusions derived from the axioms correspond to reality.

This strategy of **finite convergence** avoids the theoretical predicament of infinite regress while aligning with actual scientific practice—any theory stops asking questions at some depth, leaving more fundamental presuppositions to be tested by practice and validated by history.

3.5 Summary

This chapter has constructed a four-principle convergence mechanism for the philosophy-to-science transition, grounded in the axiomatic system of *System Ontology*:

- Defined philosophy as the potentiality field Φ —the set of all logically possible system laws;
- Defined science as the truth subset $\Sigma = \{x \in \Phi \mid \text{Valid}(x)\}$;
- Specified $\text{Valid}(x)$ as four principles of legitimacy: logical legitimacy, observability legitimacy, formal legitimacy, and boundary legitimacy;
- Demonstrated the axiomatic sources, independence, and completeness of the four principles;
- Articulated the methodological principle of finite convergence.

This framework transforms the demarcation problem from a static “standard” into a dynamic “convergence process,” from external “rules” into intrinsic “conditions of existence.” In the next chapter, we will test this mechanism through three historical case studies—the transition from natural philosophy to classical mechanics, from alchemy to chemistry, and Darwinian evolution—demonstrating how it operates in actual scientific practice.

4. Historical Cases: How the Four Principles Screen Science

The preceding chapter derived the four-principle convergence mechanism from the axioms of *System Ontology*. This chapter tests this mechanism against concrete episodes in the history of science. We select three paradigmatic cases—the transition from natural philosophy to classical mechanics, the transition from alchemy to chemistry, and Darwinian evolution as a weakly formalized science—to demonstrate how the four principles operate in actual scientific practice. These cases span different historical periods and different levels of formalization, collectively illustrating that any body of knowledge recognized as science must pass through the screen of the four principles.

4.1 Case One: From Natural Philosophy to Classical Mechanics

4.1.1 Ancient Greece to the Middle Ages: Speculation on Motion (Philosophical Stage)

From Aristotle to medieval scholastic philosophy, thinking about “motion” belonged to the realm of philosophical speculation. Aristotle’s natural philosophy distinguished “natural motion” (such as heavy objects falling, light objects rising) from “violent motion” (such as projectile motion), holding that an object’s state of motion depends on its “natural place.” The medieval impetus theory modified this view but remained at the level of conceptual deduction.

The theories of motion in this period, though logically consistent (satisfying logical legitimacy), lacked observability legitimacy and formal legitimacy. Aristotle’s laws of motion (e.g., “the speed of a falling body is proportional to its weight”) were not systematically tested by experiment, and their expression remained within natural language and metaphor, making them impossible to operate publicly or predict precisely. They belonged to the potentiality field Φ as one possibility among many, not yet having entered Σ .

4.1.2 Galileo: The Introduction of Experiment and Mathematics (The Initial Opening of Observability and Formal Legitimacy)

Galileo’s work marked a critical turning point from philosophy toward science. Through inclined plane experiments and free-fall experiments, he systematically placed motion phenomena under controlled observational conditions and attempted to describe them in mathematical language—for example, the formula for uniformly accelerated motion $v \propto t$. His use of thought experiments (such as the legendary Leaning Tower of Pisa experiment) further strengthened observability legitimacy: even when experimental conditions could not be perfectly realized, thought experiments could still yield testable conclusions.

However, Galileo’s work remained at the semi-formalization stage: the laws of motion had not yet been integrated into an axiomatic system, and the principle of inertia and the law of free fall lacked a unified mathematical framework. The boundary legitimacy principle was also not yet explicit—Galileo did not recognize that his theory would be limited at high speeds or under strong gravity.

4.1.3 Newton: Axiomatic System and Strong Formalization (All Four Principles Satisfied)

Newton completed the final convergence of classical mechanics in *Philosophiæ Naturalis Principia Mathematica* (1687). He proposed the three laws of motion and the law of universal gravitation, formalizing them in the language of calculus into a computable axiomatic system:

- **Logical legitimacy:** The three laws are internally consistent, and Kepler’s laws of planetary motion can be derived from them without contradiction.
- **Observability legitimacy:** The laws can be repeatedly verified by astronomical observations

(planetary orbits) and terrestrial experiments (such as pendulum motion), and they predicted previously unknown phenomena (such as the return of Halley's Comet).

- **Formal legitimacy:** Precisely expressed in the language of calculus, the system can be computed identically by any trained mathematical physicist.
- **Boundary legitimacy:** Newton himself acknowledged the difficulty of defining absolute time and absolute space, implicitly recognizing the theory's limited scope; later, relativity theory and quantum mechanics would reveal the boundaries of classical mechanics (low speeds, macroscopic scales).

The formation of Newtonian mechanics exemplifies the standard operation of the philosophy-to-science convergence mechanism: the problem of motion, from philosophical speculation in antiquity, passed through experiment and mathematics, eventually collapsing into a computable, testable, publicly operable strongly formalized science.

4.2 Case Two: From Alchemy to Chemistry

4.2.1 The Alchemical Period: Metaphorical Symbols and the Absence of Formal Legitimacy

Alchemy, from the Hellenistic period into early modernity, aimed to transform base metals into noble metals and prepare elixirs of immortality. Alchemists accumulated rich experimental experience (thus relatively strong observability legitimacy), but their theoretical expression relied heavily on metaphorical symbols—such as the “philosopher's stone,” the “sulfur-mercury theory,” and the “yin-yang” of Chinese alchemy. These symbols carried ambiguous meanings across different alchemical traditions, making public operation and precise replication impossible—i.e., lacking formal legitimacy. Alchemical theories often contained internal contradictions (such as simultaneously pursuing “material transformation” and “spiritual purification”), and thus also fell short of logical legitimacy. Consequently, despite its observational basis, alchemy remained in the potentiality field and did not enter science.

4.2.2 Boyle and Lavoisier: Experimental Norms and Quantitative Methods (Enhancing Observability and Formal Legitimacy)

Robert Boyle, in *The Sceptical Chymist* (1661), systematically criticized alchemy's mystical symbols and emphasized that chemistry should be based on repeatable experiments and clear terminology. He introduced the concept of “element” (a substance that cannot be further decomposed) and standardized the format of experimental reports. This work significantly elevated chemistry's formal legitimacy: experiments could now be repeated by others, and concepts had relatively stable referents.

Antoine Lavoisier in the late eighteenth century carried out a more thorough revolution. He introduced the law of conservation of mass, replaced qualitative descriptions with quantitative balances, and expressed chemical reactions as measurable equations (e.g., $2H_2 + O_2 \rightarrow 2H_2O$). Lavoisier's oxygen theory replaced the phlogiston theory, marking chemistry's critical leap from semi-formalization to strong formalization. By this stage, chemistry satisfied logical legitimacy (conservation of mass is consistent), observability legitimacy (experiments repeatable), and formal legitimacy (quantitative equations)—though boundary legitimacy was not yet fully articulated.

4.2.3 Mendeleev: The Periodic Table and Systematization (Establishing Boundary Legitimacy)

Dmitri Mendeleev proposed the periodic law in 1869, arranging the known elements by atomic weight and chemical properties into a table, and accurately predicted the properties of undiscovered elements

(such as gallium, scandium, germanium). The periodic table not only possessed great predictive power but more importantly provided a clear boundary for chemical knowledge: the properties of elements were organized into a finite, closed system, eliminating the possibility of “infinite hybridization.” The success of the periodic table marked the completion of chemistry’s convergence through all four principles. It strongly formalized the relationships among elements and pointed toward deeper theory (quantum mechanics) at its boundary—the structure of the periodic table was ultimately explained by electron configuration, which itself is bounded by the fundamental constants of quantum mechanics. Chemistry had transformed from the chaotic possibilities of alchemy, through experimental norms, quantification, and systematization, into a mature science.

4.3 Case Three: Darwinian Evolution—A Weakly Formalized Science

4.3.1 Pre-Darwinian Period: Natural Theology and Species Fixism (Philosophical Stage)

Before Darwin, explanations of biodiversity were dominated by natural theology (species separately created by God) and species fixism. These views possessed logical consistency (satisfying logical legitimacy) but were weak in observability legitimacy (lacking systematic fossil records and biogeographic evidence) and even weaker in formal legitimacy (no testable mechanism). Linnaeus’s taxonomy provided a foundation for formal legitimacy but did not explain the origin of species. Understanding of life phenomena in this period remained within the realm of philosophical speculation.

4.3.2 Darwin: The Variation-Selection-Heredity Framework (Initial Satisfaction of the Four Principles)

Darwin proposed the theory of natural selection in *On the Origin of Species* (1859): individuals within a population vary; favorable variations increase the probability of survival and reproduction; over long periods, this leads to species divergence. The theory was expressed in natural language without mathematical formulas, yet its core concepts were clear, its logic consistent, its claims observable, and its predictions falsifiable.

We examine the four principles in turn:

- **Logical legitimacy:** The natural selection mechanism is internally consistent (variation-selection-heredity forms a closed loop) and can uniformly explain a wide range of biological phenomena (adaptation, speciation, biogeographic patterns).
- **Observability legitimacy:** At the time of proposal, there was already substantial observational evidence (morphological variation in Galápagos finches, artificial selection in breeding, progressive patterns in the fossil record). Subsequent genetics and molecular biology have provided continuous new tests.
- **Formal legitimacy:** Although lacking mathematical formulas, its core concepts (fitness, selection pressure, heritable variation) are clearly defined and can be quantified and modeled by later researchers (e.g., population genetics). Formal legitimacy is not equivalent to mathematization—Darwin’s theory possesses a publicly operable structure.
- **Boundary legitimacy:** Darwin explicitly acknowledged that the “mechanism of heredity” was then a black box (“the laws of variation are unknown”), which precisely reflects the theory’s open boundaries. Later, Mendelian genetics filled this gap, confirming the theory’s resilience and the expandability of its boundaries.

4.3.3 Implications: Formalization Is Not Equivalent to Mathematization

Darwinian evolution is a paradigm case of **weakly formalized science**. It demonstrates that a scientific theory need not contain mathematical formulas; it suffices to satisfy the four principles—logically consistent, observable, publicly operable, and boundary-aware. This case powerfully refutes the prejudice that “formalization = mathematization” and confirms the definition of formal legitimacy earlier in this paper (structurable, unambiguous expression, rather than necessarily strongly formalized).

4.4 Summary of Cases

The three cases illustrate the operation of the four principles across different historical contexts:

Case	Initial State	Convergence Process	Final Scientific Form
Natural Philosophy → Classical Mechanics	Speculation on motion (logically consistent, lacking observability and form)	Galileo introduced experiment and mathematics → Newton axiomatized	Strongly formalized science
Alchemy → Chemistry	Metaphorical symbols (observability acceptable, lacking form)	Boyle standardized terminology → Lavoisier quantified → Mendeleev systematized	Semi→strongly formalized science
Darwinian Evolution	Natural theology (logically consistent, lacking observability and form)	Darwin proposed variation-selection-heredity framework	Weakly formalized science

Common conclusion: Any body of knowledge recognized as science must pass through the screen of the four principles—logical consistency is the starting point; observability legitimacy ensures connection with experience; formal legitimacy makes knowledge publicly operable; boundary legitimacy establishes the theory’s domain of applicability. The gradual satisfaction of the four principles is the underlying mechanism by which philosophical speculation converges into scientific theory.

In the next chapter, we will further explore the three levels of formalization and the role of mathematics, clarifying the position of the present theory within the formalization spectrum.

5. Levels of Formalization and the Role of Mathematics

The preceding chapter tested the four-principle convergence mechanism through historical case studies. In those cases, we saw that classical mechanics moved toward strong formalization (mathematical axiomatic system), chemistry underwent a transition from semi-formalization to strong formalization, while Darwinian evolution persisted as a weakly formalized science for an extended period. This raises a fundamental question: **What is formalization? What is its relationship to mathematics? Must science be mathematized?**

This chapter aims to clarify the three levels of formalization, to show that “mixed formalization” is the normal state of theoretical evolution, and to situate the role of mathematics in science—it is the primary tool for strong formalization and thus a **sufficient but not necessary condition** for science. This framework provides a unified perspective for understanding the different scientific forms of various disciplines and simultaneously grounds the self-positioning of the present theory.

5.1 Three Levels of Formalization

Formalization refers to the process of transforming knowledge from a state dependent on intuition, natural language, and individual experience into a form characterized by explicit symbols, fixed structures, unambiguous rules, and mechanical execution. Depending on the degree of formalization, theories can be classified into three levels:

5.1.1 Weak Formalization

Characteristics: Uses natural language but possesses clear definitions, explicit structure, discernible rules, and criteria for judgment. No specialized symbolic system is introduced, but logical relationships are transparent and reasoning traceable.

Examples:

- **Darwinian evolution:** The “variation-selection-heredity” framework is articulated in natural language, yet concepts such as “fitness” and “selection pressure” have clear meanings, and the theoretical structure is discernible.
- **Linnaean taxonomy:** Uses Latin binomial nomenclature to name organisms; though it lacks formulas, the classificatory rules are explicit and publicly operable.
- **Plate tectonics in geology:** Initially described continental drift and seafloor spreading in natural language, yet core concepts (plates, boundaries, direction of motion) are clearly defined.

Applications: Initial stages of new fields, preliminary convergence of philosophical speculation, initial communication across disciplines. Weakly formalized theories satisfy the four principles—as long as they are logically consistent, observable, publicly operable, and boundary-aware—and are therefore scientific.

5.1.2 Semi-Formalization

Characteristics: Introduces specialized symbolic systems (such as mathematical or logical symbols), with some relationships expressible in formulas, though not yet fully computable. Certain parts of the theory still rely on natural language explication, but the core logic has been symbolized.

Examples:

- **Supply-demand model in economics:** $Q_d = f(P)$, $Q_s = g(P)$, equilibrium condition $Q_d = Q_s$ expressed symbolically, while market structure, preferences, etc., still rely on natural language.
- **Formulas in System Ontology:** such as $C_{\text{dimension}}(D) = c_0 + c_1 \cdot D^\alpha$ (dimensional maintenance cost)—core relationships symbolized, though the overall theory remains primarily in natural language.
- **Early chemistry:** Lavoisier used chemical equations to represent reactions, but atomic structure and chemical bonds had not yet been symbolized.

Applications: Mid-stage development of a theory, when core mechanisms have been identified and

require precise expression for derivation and application. Semi-formalization is the crucial transitional phase from weak to strong formalization.

5.1.3 Strong Formalization

Characteristics: The theory is expressed entirely in mathematical language; all concepts have strict definitions; all reasoning can be formalized as computation or proof. The theory is computable, predictable, and even programmable.

Examples:

- **Newtonian mechanics:** $F = ma$, universal gravitation $F = G \frac{m_1 m_2}{r^2}$; equations of motion can be solved.
- **Maxwell's equations:** Unified electromagnetic phenomena using partial differential equations, enabling precise calculation of electromagnetic wave propagation.
- **Quantum mechanics:** Schrödinger equation $i\hbar \frac{\partial}{\partial t} \Psi = \hat{H} \Psi$; energy levels and probability amplitudes can be computed.

Applications: Mature stages of a theory, suitable for engineering design and precise prediction. Strong formalization is not the endpoint for all sciences—many disciplines (such as evolutionary biology, climate science) exhibit strong formalization in localized subfields while remaining semi-formalized overall.

5.2 Mixed Formalization as the Normal State of Theoretical Evolution

The formalization level of a theory as a whole is determined by its primary mode of expression and its core chain of reasoning, but **localized occurrences of higher-level symbolization do not contradict the theory's overall level**. Mixed formalization is the normal state of theoretical evolution.

5.2.1 Mixed Formalization within Disciplines

- **Evolutionary biology:** The core framework (natural selection) is weakly formalized, but population genetics employs strongly formalized equations such as the Hardy–Weinberg equilibrium $p^2 + 2pq + q^2 = 1$, and molecular evolution uses substitution rate models.
- **Climate science:** Global circulation models (GCMs) are strongly formalized (numerically solving fluid dynamics equations), while assessments of climate sensitivity still rely on semi-formalized conceptual models.
- **Cognitive science:** Subfields such as psychophysics use strong formalization (Stevens' power law), while consciousness research remains at the stage of weak formalization.

5.2.2 The Natural Path of Theoretical Evolution

A new theory often begins with weak formalization. After core concepts and frameworks are established, symbolic expression is gradually introduced, and some subfields move toward strong formalization. This mixed state is not a defect but a sign of theoretical vitality: it preserves the flexibility and comprehensibility of the core framework while achieving precision in quantifiable domains.

Position of the present theory: The philosophy-to-science mechanism constructed in this paper is currently at the **semi-formalization** stage. We have introduced symbolic formulas (such as $\Sigma = \{x \in$

$\Phi \mid \text{Valid}(x)\}$ and the definitions of the four principles), but much of the exposition still relies on natural language, and the reasoning has not been fully formalized. Core concepts (such as “self-organizing tendency,” “systemic constraint”) are clearly defined but have not yet been quantified as computable variables. This semi-formalized state is appropriate—the task of this paper is to provide a meta-theoretical framework for “how philosophy becomes science,” not to perform concrete engineering calculations. Its strength lies in explanatory power and structural clarity, not predictive precision. In the future, when applied to specific fields (such as cognitive science or the sociology of science), it may be further developed into strong formalization.

5.3 Mathematics: A Sufficient but Not Necessary Condition for Science

5.3.1 The Role of Mathematics in Science

Mathematics is the primary tool for strong formalization, and its roles in science include:

- **Precision:** Eliminates the ambiguity of natural language, making reasoning mechanically executable.
- **Computability:** Enables prediction, simulation, and engineering design.
- **Unification:** Different phenomena may share the same mathematical structure (e.g., the wave equation applies to sound, light, and water waves).

However, mathematics is not the only form of science, nor is it the core criterion for science.

5.3.2 Why Mathematics Is Not a Necessary Condition

The historical cases presented earlier already demonstrate this:

- Darwinian evolution is universally recognized as science despite lacking mathematical formulas.
- Plate tectonics and the central dogma of molecular biology ($\text{DNA} \rightarrow \text{RNA} \rightarrow \text{protein}$) were initially articulated in natural language yet transformed their respective fields.
- Qualitative research in the social sciences (e.g., ethnographic fieldwork) and historical reasoning also follow the four principles and qualify as weakly formalized science.

These cases collectively show: **a theory is scientific as long as it satisfies the four principles—logically consistent, observable, publicly operable, and boundary-aware—regardless of whether it contains mathematical formulas.**

5.3.3 Why Mathematics Is a Sufficient Condition

If a theory achieves strong formalization (i.e., is fully mathematized), it automatically satisfies the four principles:

- Mathematical theories require internal consistency (logical legitimacy).
- Mathematized theories typically correspond to observable phenomena (observability legitimacy); otherwise, they would be pure mathematics rather than science.
- Mathematical expression is inherently publicly operable (formal legitimacy).
- Mathematical theories usually specify their axioms and domain of definition (boundary legitimacy).

Thus, mathematization is a **sufficient condition** for science: it guarantees high precision, computability, and predictability. But it is not a **necessary condition**—weakly formalized and semi-formalized sciences are equally legitimate.

5.3.4 Response to the “Mathematical Hegemony” Bias

There is often an implicit bias in contemporary science that only mathematized theories count as “hard science,” while qualitative research is “soft science.” This bias confuses “formalization” with “mathematization,” overlooking the fact that the core of science is a **publicly testable cognitive framework**, not the mode of expression. The four-principle mechanism proposed in this paper provides a more inclusive framework: any body of knowledge that satisfies logical, observability, formal, and boundary legitimacy should be recognized as science. Mathematics is the jewel in the crown of science, but the foundation of the crown is composed of weakly formalized and semi-formalized sciences.

5.4 Summary

This chapter has clarified the three levels of formalization—weak, semi, strong—along with their characteristics and typical applications. It has shown that mixed formalization is the normal state of theoretical evolution, and has articulated the role of mathematics in science: it is the primary tool for strong formalization and therefore a **sufficient but not necessary condition** for science. This framework provides a unified perspective for understanding the scientific forms of different disciplines and offers a theoretical justification for why Darwinian evolution, as a weakly formalized science, is nonetheless legitimate.

In the next chapter, we will complete the self-referential closure of the meta-theory: we will test whether the theory proposed in this paper—the theory of how philosophy becomes science—satisfies the four principles itself, and demonstrate how such self-referential legitimacy avoids vicious circularity.

6. Self-Referential Closure of the Meta-Theory

The preceding chapters derived the four-principle convergence mechanism from the axioms of *System Ontology*, tested it through historical case studies, and clarified the three levels of formalization and the role of mathematics. At this point, a fundamental question emerges: **Must the meta-theory—the theory of how science becomes science—itself satisfy the four principles? If so, can it pass its own test? And if it passes, does this not constitute a vicious circle (proving itself by itself)?**

The task of this chapter is to complete the **self-referential closure** of this meta-theory. We will first argue that the meta-theory must undergo examination by the four principles, then test the present theoretical framework against each principle, and finally demonstrate how the **external judgment stance** makes self-reference a benign closure rather than a vicious circle. The completion of this closure grants the present theory full qualification as **meta-science**—it not only explains the world but also explains why it itself is capable of explaining the world.

6.1 The Meta-Theory Must Undergo Examination by the Four Principles

If the four principles proposed in this paper are the legitimate criteria for scientific admission, then any theory claiming to be about “how science becomes science”—that is, any meta-scientific theory—must also be examined by these criteria. Otherwise, we would fall into a double standard: the four principles apply to all sciences except the very theory that discusses science itself. Such exceptionalism would undermine the theory’s claim to universality.

Therefore, the meta-theory must satisfy:

1. **Logical Legitimacy:** Internally consistent, free from self-contradiction.
2. **Observability Legitimacy:** Capable of interacting with the empirical world and leaving detectable effects.
3. **Formal Legitimacy:** Clearly expressible and transmissible, publicly operable.
4. **Boundary Legitimacy:** Acknowledges its own domain of applicability and incompleteness.

These are the prerequisites for the meta-theory to acquire “existential legitimacy” (see *System Ontology*, Meta-Axiom 1).

6.2 Self-Examination: The Present Theory Passes the Four Principles

Let us test the theoretical framework constructed in this paper against each principle in turn.

6.2.1 Logical Legitimacy

The argument of this paper proceeds deductively from the axioms of *System Ontology*:

- Meta-Axiom 0 (the principle of self-organization) defines the potentiality field Φ .
- Meta-Axiom 1 (the principle of systemic constraint) provides the ultimate source of legitimacy screening.
- Axioms 2, 3, and 5, together with Theorem 2, yield formal legitimacy, observability legitimacy, and boundary legitimacy respectively.
- The axiomatic system is internally consistent, with no contradictions introduced throughout.

The demarcation between philosophy and science is clear, and the four principles are independent and complete. **Logical legitimacy is satisfied.**

6.2.2 Observability Legitimacy

The present theory is not purely speculative:

- It provides operational criteria (the four principles) for scientific practice, directly guiding researchers in determining what merits investigation and what qualifies as science.
- It is corroborated by historical case studies (Chapter 4), which show that the four principles actually operated in the formation of classical mechanics, chemistry, and evolutionary theory.
- It can be applied to contemporary scientific controversies (e.g., “Is artificial intelligence scientific?” “Is consciousness research scientific?”) yielding testable predictions.
- Thus, the theory maintains a codifiable interaction channel with the empirical world. **Observability legitimacy is satisfied.**

6.2.3 Formal Legitimacy

This paper employs semi-formalized expression:

- Set-theoretic symbols define Φ and Σ .
- Core relationships are expressed in formulas (e.g., $C_{\text{dimension}}(D) = c_0 + c_1 \cdot D^\alpha$, $\Sigma = \{x \in \Phi \mid \text{Valid}(x)\}$).
- The structure is clear (eight progressive chapters), and core concepts (the four principles, levels of formalization) are explicitly defined.
- It can be publicly read, understood, criticized, and applied.

Thus, **formal legitimacy is satisfied.**

6.2.4 Boundary Legitimacy

The present theory explicitly acknowledges its own boundaries and incompleteness:

- **Logical level:** This paper acknowledges that any sufficiently complex self-referential system cannot prove its own completeness within itself (the application of Gödel's theorems at the meta-level). Therefore, it does not claim to have “proved its own absolute truth”; it only claims to provide a “superior framework.”
- **Empirical level:** The paper clearly delineates its domain of applicability—explaining the underlying mechanism of “how philosophy becomes science,” rather than providing experimental designs or engineering solutions. For more specific empirical questions (such as precise quantification of the four principles, determination of phase transition thresholds), it leaves open interfaces for future research.

Boundary legitimacy is satisfied.

6.3 The External Judgment Stance: Avoiding Vicious Circularity

One might object: You use the four principles you yourself proposed to test yourself—isn't this circular reasoning?

This objection touches the core predicament of self-referential theories. Our response is: **The self-examination is not performed by the theory itself, but by the reader/author at the meta-level, applying the four principles to the theory as an object.**

6.3.1 The Lesson of Gödel's Theorems

Gödel's incompleteness theorems tell us that any sufficiently powerful formal system cannot prove its own consistency within itself. But this does not mean the system's consistency is unknowable—it can be “seen” from outside the system, i.e., proven at the meta-level.

By analogy, the self-examination of the present theory is not a case of “the theory proving itself,” but of the reader (or author) standing outside the theory, treating the theory as an object, and applying the four principles to make a judgment. This judgment is performed at the **meta-level**, not as a self-reference within the theory itself.

6.3.2 The Operation of External Judgment

In the writing and reading of this paper:

- **The author**, while constructing the theory, consciously applied the four principles to examine it (as shown in Section 6.2).
- **The reader**, while reading, can similarly apply the four principles independently to judge whether the theory satisfies the criteria for scientific admission.
- This **external judgment stance** is the only legitimate path for any self-referential system to avoid paradox—just as the truth of Gödel sentences is “seen” from outside the system rather than proven within it.

Therefore, the self-referential closure of this paper is **benign**: it is not a closed self-loop but an open, publicly examinable meta-level judgment.

6.4 Positioning the Theory as a Cognitive Scalpel

Having completed self-examination, the present theory qualifies as **meta-science**. Yet it does not claim absolute truth—this is precisely an expression of boundary legitimacy.

6.4.1 Does Not Claim Absolute Truth

This theory acknowledges:

- It is only one among many possible meta-theoretical frameworks (Axiom 4: the relativity of cognitive frameworks).
- It cannot prove its own completeness (Gödel's theorems).
- It may be revised or replaced by superior frameworks in the future (the openness of theoretical evolution).

This self-awareness is precisely the source of its legitimacy: a theory that claims to “explain everything” would lose legitimacy by attempting to transcend boundaries; a theory that acknowledges its own boundaries and incompleteness gains stable existential legitimacy.

6.4.2 As a “Cognitive Scalpel”

This paper positions itself as a **cognitive scalpel**—a sharp, operable, refinable analytical tool, rather than a statue of truth to be worshiped.

Its value lies in:

- **Explanatory power:** It can uniformly explain the demarcation and convergence phenomena in the history of science.
- **Operationality:** The four principles can be directly applied to assess contemporary scientific controversies.
- **Openness:** It leaves interfaces for future empirical research and interdisciplinary dialogue.

6.5 Summary

This chapter has completed the self-referential closure of the meta-theory:

1. **The meta-theory must undergo examination by the four principles;** otherwise, it would fall into a double standard.
2. **The examination is passed:** The present theory satisfies logical legitimacy, observability legitimacy, formal legitimacy, and boundary legitimacy.
3. **The external judgment stance:** The self-examination is not performed by the theory itself but by the reader/author at the meta-level, applying the four principles to make an external judgment. This avoids vicious circularity.
4. **Positioning:** The theory does not claim absolute truth but positions itself as a publicly examinable, iteratively refinable “cognitive scalpel.”

Thus, the philosophy-to-science formal mechanism constructed in this paper not only explains the sciences of others but also explains itself. It has achieved the transition from “description” to “mechanism” and has thereby proven its own legitimacy as meta-science.

In the next chapter, we will systematically engage with the traditions of Popper, Kuhn, Lakatos, Feyerabend, and Gödel, demonstrating how the present theory integrates and transcends them.

7. Dialogue with the Traditions of Philosophy of Science

The preceding chapters derived the four-principle convergence mechanism from the axioms of *System Ontology*, tested it through historical case studies, clarified the levels of formalization and the role of mathematics, and completed the self-referential closure of the meta-theory. The theoretical framework

of this paper is now fully articulated. Yet the value of any new theory lies not only in its internal coherence but also in its relationship to the existing body of knowledge—whether it can absorb the insights of its predecessors, transcend their limitations, and establish its own academic position through dialogue.

This chapter systematically engages with the core traditions of the demarcation problem in philosophy of science: Popper’s falsificationism, Kuhn’s paradigm theory, Lakatos’s methodology of scientific research programmes, Feyerabend’s epistemological anarchism, and the philosophical implications of Gödel’s incompleteness theorems. We will show how the four-principle mechanism proposed in this paper absorbs, integrates, and transcends these traditions, ultimately achieving the transition from “description” to “mechanism.”

7.1 Integrating and Transcending Popper’s Falsificationism

7.1.1 Absorption: Falsification as an Important Manifestation of Observability Legitimacy

The core insight of Popper’s falsificationism is that scientific theories must rule out certain possible observations, thereby accepting the risk of empirical refutation. This insight is absorbed into the **observability legitimacy** principle of this paper. Observability legitimacy requires that scientific propositions maintain a principled codifiable interaction channel with the cognitive framework—meaning they must be testable, and the most direct form of testability is precisely falsifiability. Thus, Popper’s falsificationism is not discarded but incorporated into a broader condition. A proposition that satisfies observability legitimacy can in principle be empirically tested—the test may take the form of confirmation, falsification, or probabilistic updating, with falsification being the most stringent of these.

7.1.2 Transcendence: Falsification as a Sufficient but Not Necessary Condition

However, falsificationism cannot account for two important phenomena in the history of science:

- **The initial unfalsifiability of new theories:** Darwinian evolution, at the time of its proposal, lacked direct transitional fossil evidence, yet it was recognized as scientific. Popper himself once characterized evolution as a “metaphysical research programme,” a view at odds with the consensus of scientific history.
- **The resilience of theories in the face of anomalies:** A theory is not abandoned after a single counterexample; scientists adjust auxiliary hypotheses or refine measurement methods.

Popper’s “instant falsification” does not reflect actual scientific practice.

Observability legitimacy as defined in this paper is broader than falsifiability: it only requires that the theory be capable in principle of interacting with the empirical world, not that such interaction must be falsifying. Darwinian evolution already satisfied observability legitimacy in Darwin’s time (Galápagos finch beak variation, artificial selection in breeding), even though it had not yet been strictly falsified. This inclusiveness better aligns with the reality of scientific practice.

7.2 Reinterpreting Kuhn’s Paradigm Theory

7.2.1 Absorption: Paradigm as Cognitive Framework

Kuhn’s paradigm theory revealed the social structure and historical dynamics of science: normal science is governed by a paradigm; paradigm shifts are revolutionary. This paper absorbs this insight into a more fundamental cognitive framework theory—a **paradigm** is precisely a **cognitive framework** as defined in *System Ontology* Axiom 5: a set of boundary-defining, information-filtering,

internal-process-constraining basic presuppositions.

A paradigm shift is thus a **cognitive escape**—a leap of cognitive framework. The present framework provides a dynamical explanation for this: when the old framework can no longer accommodate accumulated anomalies (the build-up of structural tension), and a new framework performs better on the four principles (logical consistency, observability, formal expressibility, boundary clarity), cognitive escape occurs.

7.2.2 Transcendence: From Description to Mechanism, From Relativism to Objective Criteria

Kuhn's contribution was **descriptive**: he showed that paradigm revolutions happen, but he did not explain **why they must happen** or **on what grounds a new paradigm replaces the old**. His account of “incommensurability” even carried a relativistic undertone—different paradigms cannot be rationally compared.

The four principles of this paper provide **operational objective criteria** for paradigm change:

- Does the new paradigm eliminate logical contradictions present in the old? (logical legitimacy)
- Can the new paradigm explain observational phenomena that the old could not? (observability legitimacy)
- Does the new paradigm provide a clearer, more operable conceptual framework? (formal legitimacy)
- Does the new paradigm explicitly define its own domain of applicability, avoiding the old paradigm's infinite nesting? (boundary legitimacy)

These criteria are independent of any particular scientific community's consensus, thus avoiding relativism. One paradigm replaces another not because the community votes for it, but because it performs better on the four principles.

7.3 Formalizing Lakatos's Research Programmes

7.3.1 Absorption: The Hard Core and Protective Belt Structure

Lakatos conceived the development of theories as the evolution of **scientific research programmes**: each programme consists of an inviolable **hard core** and an adjustable **protective belt**. Scientists adjust the protective belt to cope with anomalies rather than immediately abandoning the hard core. This structure is preserved in the present framework: the hard core corresponds to the core presuppositions of a cognitive framework (e.g., Newton's three laws of motion), while the protective belt corresponds to auxiliary hypotheses and formal expressions (e.g., perturbation calculations for planetary orbits).

7.3.2 Transcendence: From Retrospective Evaluation to Prospective Criteria, From Static Structure to Dynamic Mechanism

Lakatos's limitations are twofold:

- His judgments of “progressive” versus “degenerating” rely on retrospective historical evaluation, lacking predictive power that could be applied prospectively.
- He did not explain how the hard core itself acquires legitimacy, nor did he specify the driving force behind programme evolution.

The four principles of this paper provide **prospective criteria** for research programmes: does the new programme satisfy the four principles? Does it perform better than the old programme on these criteria?

Such judgments can be applied at early stages of a programme's development, not merely as post-hoc evaluations.

Moreover, the **structural tension as driver of evolution** theorem (*System Ontology*, Theorem 4) provides a dynamical mechanism for programme evolution: tensions between the hard core and the protective belt, and between the programme and its environment, drive adjustments to the protective belt and, when tensions accumulate beyond the framework's capacity, either programme collapse or cognitive escape to a new programme.

7.4 Responding to Feyerabend

7.4.1 Absorption: The Legitimacy of Methodological Pluralism

Feyerabend, in *Against Method*, criticized all forms of methodological dogmatism, arguing that “anything goes.” This position reveals the diversity and creativity inherent in scientific practice—scientific breakthroughs often occur when rules are broken. This paper fully endorses this insight: there is no single, eternal “scientific method.”

7.4.2 Response: The Four Principles Are Conditions of Existence, Not Methods

However, Feyerabend's relativism goes to an extreme: if “anything goes,” then no distinction can be made between science and pseudoscience, depriving science education, research funding, and public policy of a rational basis.

The response of this paper is: **the four principles are not “methods,” but “conditions of existence.”** They do not prescribe how science should be produced (methods may be plural), but only the minimal conditions that scientific knowledge must satisfy. Any body of knowledge that passes the four principles, regardless of the method by which it was produced (speculation, experiment, simulation, intuition), qualifies as science. Conversely, any body of knowledge that fails the four principles, regardless of its method, cannot be science.

This position respects methodological pluralism while providing a clear boundary for science, thereby avoiding relativism.

7.5 Reinterpreting Gödel's Incompleteness Theorems

7.5.1 Traditional Interpretation: Incompleteness as a Defect

Gödel's incompleteness theorems are often interpreted as revealing a “defect” in mathematics or formal systems: any sufficiently powerful formal system is either incomplete or inconsistent. This interpretation has reinforced the fear that “self-reference inevitably leads to paradox,” causing philosophy of science to long avoid the self-referential problems of meta-theory.

7.5.2 The Present Reinterpretation: Incompleteness as a License to Exist

Within the framework of *System Ontology*, we offer a different interpretation: **incompleteness is not a defect but a condition of existence for any complex self-referential system.**

Meta-Axiom 1 (the principle of systemic constraint) stipulates that any stably existing system must contain within itself a set of self-constraining meta-rules that eliminate the “vicious self-reference” which would lead to logical collapse. A system that attempts to achieve “completeness and consistency” is precisely engaged in vicious self-reference—it tries to be its own ultimate judge. Meta-Axiom 1 prevents the system from falling into a logical black hole by introducing structural incompleteness.

Thus, the “undecidable propositions” revealed by Gödel’s theorems are not scars on mathematics but **immune sentinels** of mathematical systems. Incompleteness is the **license to exist** for any complex self-referential system.

7.5.3 Application to the Meta-Theory

The present theory acknowledges its own incompleteness (Chapter 6) and transforms this incompleteness into a driving force for theoretical evolution—it does not seek a once-and-for-all “absolute truth” but positions itself as a refinable cognitive scalpel, honed through application. This self-awareness is the deepest resonance with Gödel’s theorems.

7.6 Summary

This chapter has systematically engaged with the core traditions of the demarcation problem in philosophy of science:

Tradition	Absorption	Transcendence
Popper’s falsificationism	Falsification as an important manifestation of observability legitimacy	Falsification is only a sufficient condition; observability legitimacy is broader
Kuhn’s paradigm theory	Paradigm as cognitive framework	From description to mechanism, from relativism to objective criteria
Lakatos’s research programmes	Hard core / protective belt structure	From retrospective evaluation to prospective criteria, from static structure to dynamic mechanism
Feyerabend’s methodological pluralism	The legitimacy of methodological diversity	The four principles are conditions of existence, not methods, avoiding relativism
Gödel’s theorems	The warning of self-referential paradox	Incompleteness as the license to exist for complex self-referential systems

This dialogue shows that the four-principle mechanism proposed in this paper is not a negation of these traditions but an integration and transcendence of their insights—elevating description to mechanism, demarcation to convergence, static criteria to dynamic screening.

In the final chapter (Chapter 8), we will summarize the main contributions of this paper and point to directions for future research.

8. Conclusion and Outlook

8.1 Main Contributions

This paper has constructed a formal underlying mechanism for how philosophy becomes science, grounded in the axiomatic system of *System Ontology*, and has completed the self-referential closure of

the meta-theory. The main contributions can be summarized in five respects:

First, a reconceptualization of the relationship between philosophy and science.

We have defined philosophy as the *potentiality field* Φ —the set of all logically possible system laws, guaranteed by Meta-Axiom 0 (the principle of self-organization)—and science as the *truth subset* $\Sigma = \{x \in \Phi \mid \text{Valid}(x)\}$ selected from this field. This set-theoretic definition transforms the philosophy–science relationship from a static opposition or temporal succession into a dynamic relation between “space of possibility” and “convergent truth.” Philosophy is not a “pre-scientific stage” but the condition that makes science possible—without the unfolding of the potentiality field, there would be no candidates to screen.

Second, the derivation of four principles of legitimacy for scientific admission.

From the axioms of *System Ontology*, we derived four principles:

- **Logical legitimacy** (derived from Meta-Axiom 1): a scientific theory must be internally consistent.
- **Observability legitimacy** (derived from Axiom 3): a scientific proposition must maintain a principled codifiable interaction channel with the cognitive framework.
- **Formal legitimacy** (derived from Axiom 2): scientific knowledge must be structurable and unambiguously expressible.
- **Boundary legitimacy** (derived from Axiom 5 and Theorem 2): a scientific theory must acknowledge its own domain of applicability and avoid illegitimate infinite nesting or absolute completeness.

These four principles constitute the **convergence mechanism** from philosophy to science—the existential conditions that any body of knowledge must satisfy to qualify as science.

Third, historical case studies and the clarification of formalization levels.

Through three historical cases—the transition from natural philosophy to classical mechanics, from alchemy to chemistry, and Darwinian evolution as a weakly formalized science—we demonstrated how the four principles operate in actual scientific practice. Building on these cases, we articulated the three levels of formalization (weak, semi, strong) and argued that mathematics is a primary tool for strong formalization and thus a **sufficient but not necessary condition** for science. This framework provides a unified perspective for understanding the diverse scientific forms of different disciplines, from the strongly formalized physics to the weakly formalized evolutionary biology.

Fourth, the self-referential closure of the meta-theory.

The meta-theory proposed in this paper—the theory of how philosophy becomes science—was itself examined by the four principles and passed. Crucially, this self-examination is not a vicious circle but a benign self-referential closure achieved through an **external judgment stance**: the author and reader, standing at the meta-level, apply the four principles to the theory as an object. This stance is the only legitimate way for any self-referential system to avoid paradox, just as the truth of Gödel sentences is “seen” from outside the system rather than proven within it. The completion of this closure grants the present theory the full status of **meta-science**—it not only explains the sciences of others but also explains its own legitimacy.

Fifth, integration and transcendence of the traditions of philosophy of science.

We systematically engaged with Popper’s falsificationism, Kuhn’s paradigm theory, Lakatos’s methodology of scientific research programmes, Feyerabend’s epistemological anarchism, and the philosophical implications of Gödel’s theorems. In each case, we absorbed the core insights while transcending their limitations: elevating description to mechanism, demarcation to convergence,

external rules to internal conditions, and avoidance of self-reference to accommodation of self-reference.

8.2 Interface with *Noetic Ecology* and *Civilization Dynamics*

The philosophy-to-science formal mechanism constructed in this paper serves as the meta-theoretical foundation for *Noetic Ecology* and *Civilization Dynamics*. Together, the three works form a layered theoretical system:

- **System Ontology** (foundational layer): Provides the ultimate principles of being, defining the basic constitution and dynamics of systems. The four principles of this paper are derived from its axioms.
- **Noetic Ecology** (middle layer): Investigates the laws of noetic systems at the cognitive, cultural, and social levels. The “observability legitimacy” and “formal legitimacy” of this paper are concretized in *Noetic Ecology* as “the relativity of cognitive frameworks” and “symbols as noetic genes”; “boundary legitimacy” appears there as the lower bound $C_{\min}$ and hardware upper bound in the “cognitive expansion theorem.” The meta-cognitive power M is formalized in *Noetic Ecology* as $M = T \times C \times R$ (framework-switching capacity $\times$ logical closure strength $\times$ self-reflective frequency), providing a micro-level basis for the “interaction channel with the cognitive framework” in observability legitimacy and a dynamic explanation for the finiteness of cognitive dimensions in boundary legitimacy.
- **Civilization Dynamics** (top layer): Applies noetic ecology to the civilizational scale. The “four-principle convergence mechanism” is concretized there as the “civilizational phase transition model”—when bodies of knowledge that pass the four principles reach a critical density in society, civilization undergoes a phase transition from “collective mind” to “meta-cognitive collective intelligence.”

Thus, this paper is not only an independent meta-theoretical study but also the meta-methodological foundation of the entire “Noetic Theory Trilogy.”

8.3 Directions for Future Research

The theoretical framework of this paper still leaves several areas for future research to fill and deepen:

8.3.1 Refinement and Quantification of the Four Principles

Currently, the application of the four principles relies on qualitative analysis. Future research could attempt:

- Developing formal contradiction-detection tools (such as automated reasoning engines) for **logical legitimacy**.
- Designing operational “observability metrics” for **observability legitimacy**, distinguishing direct observation, indirect observation, and principled observability.
- Establishing a “degree of formalization index” for **formal legitimacy** to assess a theory’s position on the weak–semi–strong spectrum.
- Developing “boundary identification protocols” for **boundary legitimacy** to help researchers explicitly define the domain of applicability of their theories.

8.3.2 Application of the Four Principles to Contemporary Scientific Controversies

The four principles can be used to diagnose controversial fields in contemporary science:

- **Is artificial intelligence scientific?** Do current large language models satisfy formal legitimacy (publicly operable)? Do they satisfy observability legitimacy (testable)? Do they satisfy boundary legitimacy (acknowledging their own limitations)?
- **Is consciousness research scientific?** How do theories of consciousness (e.g., global workspace theory, integrated information theory) perform on the four principles? Which aspects still need development?
- **Qualitative research in the social sciences:** How can the four principles distinguish scientific qualitative research from pseudoscience?

8.3.3 Dialogue between the Four Principles and the Sociology of Science

The four principles are epistemological criteria, but the actual practice of science is deeply influenced by social factors. Future research could explore how the four principles interact with social institutions (such as scientific communities, peer review, resource allocation). Why is a theory that satisfies the four principles sometimes socially rejected? Why is a theory that fails the four principles sometimes socially accepted?

8.3.4 Empirical Testing of Phase Transition Thresholds

Civilization Dynamics proposes that when the proportion of awakened individuals (second-order noetic persons) in a society exceeds 21%, a civilizational phase transition may occur. What is the relationship between this threshold and the four principles? Does phase transition occur when bodies of knowledge that pass the four principles reach a critical density in society? This calls for interdisciplinary research in historical sociology and computational social science.

8.3.5 Cross-Civilizational Comparative Studies

Do the four principles apply to knowledge production traditions outside the West? Do Chinese *gewu zhizhi* (investigation of things), Indian *hetuvidyā* (logic), and the Islamic House of Wisdom movement also follow the convergence mechanism of the four principles? Cross-civilizational comparative studies would test the universality of the four principles.

8.4 Final Declaration

This paper began with a question: how does philosophical speculation converge into scientific theory? Traditional philosophy of science gave descriptions but not mechanisms. We have attempted to fill this gap.

Our answer is: **Philosophy is the unfolding of the space of possibility; science is the legitimate convergence of possibility. The mechanism of convergence is the screening by the four principles of legitimacy—logical, observability, formal, and boundary.**

This definition transforms the demarcation problem from a static “standard” into a dynamic “process,” from external “rules” into intrinsic “conditions of existence.” It explains the past and points toward the future.

Yet this paper does not claim absolute truth. It is a **cognitive scalpel**—sharp, operable, refinable. Its value lies not in its eternity but in the dialogue it opens, which is far more than it concludes.

As Gödel’s theorems remind us: any sufficiently powerful self-referential system cannot prove its own completeness within itself. This paper candidly acknowledges this limitation and takes it as proof of its own legitimacy—not because it is perfect, but because it is self-aware.

May this scalpel contribute a modest yet solid foundation for understanding science, designing

institutions, and evolving civilization.

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Glossary of Key Terms

A

Axiom

In *System Ontology*, the six system axioms (Axioms 1–6) define the necessary conditions for any system to exist: logical self-reference, material/noetic duality, interaction, value-drive, framework constraint, and hierarchical duality. The four principles of legitimacy in this paper are derived from these axioms.

B

Boundary Legitimacy

One of the four principles of legitimacy. A scientific theory must acknowledge its own domain of applicability and avoid illegitimate infinite nesting or absolute completeness. It has two levels: a logical level requirement (no infinite nesting, derived from Theorem 2) and an empirical level requirement (actual upper and lower bounds in application, such as the energy scales at which physical laws hold). It ensures that science accepts its own finitude as a condition of stable existence.

C

Cognitive Escape (*renzhi taoyi*)

A concept from *Noetic Ecology* (Theorem D23): when an individual's meta-cognitive power exceeds the gravitational pull of a dominant paradigm, they can achieve a leap in cognitive framework. In this paper, scientific revolutions are interpreted as cognitive escapes at the civilizational scale.

Cognitive Framework (*renzhi kuangjia*)

Defined in *System Ontology* Axiom 5 as the boundary, rules, and core algorithms that define a system, filter interactions, and guide internal processes. In this paper, paradigms, worldviews, and research programmes are all specific forms of cognitive frameworks. The four principles are the criteria for screening cognitive frameworks for scientific legitimacy.

Cognitive Scalpel (*renzhi shoudao*)

A metaphor used in this paper to describe the theoretical framework itself: a sharp, operable, refinable analytical tool for dissecting problems of demarcation and convergence, rather than a statue of absolute truth to be worshiped.

Convergence Mechanism (*shoulian jizhi*)

The core concept of this paper: the dynamic process by which philosophical possibilities (the potentiality field Φ) are screened by the four principles to become scientific truth (the truth subset Σ). It transforms the demarcation problem from a static standard into a dynamic operation.

D

Demarcation Problem (*huaqie wenti*)

The central question in philosophy of science: how to distinguish science from non-science (metaphysics, pseudoscience). This paper reconceptualizes it as a dynamic convergence process rather than a static boundary.

E

External Judgment Stance (*waibu panding lichang*)

The methodological stance adopted for the self-referential closure of the meta-theory: the self-examination is not performed by the theory itself but by the reader/author at the meta-level, applying the four principles to the theory as an object. This is the only legitimate way for any self-referential system to avoid vicious circularity, analogous to how the truth of Gödel sentences is “seen” from outside the system rather than proven within it.

F

Falsificationism (*zhengwei zhuyi*)

Popper’s philosophy of science, which holds that scientific propositions must be empirically falsifiable. This paper absorbs the testability insight as part of observability legitimacy but argues that falsifiability is only a sufficient condition, not a necessary one.

Finite Convergence (*youxian shoulian*)

The methodological principle of this paper: the deduction of the four principles terminates at the axioms of *System Ontology*, and the validity of those axioms is anchored in historical case studies rather than pursued in infinite regress. This avoids the theoretical predicament of infinite regress while aligning with actual scientific practice.

Formal Legitimacy

One of the four principles of legitimacy. Scientific knowledge must be structurable and unambiguously expressible using symbolic systems (language, mathematics, diagrams), and must be publicly operable. Derived from *System Ontology* Axiom 2 (material/noetic duality). Formal legitimacy is not equivalent to mathematization—weakly formalized science is equally legitimate.

Formalization, Three Levels of (*xingshihua san cengji*)

Weak formalization (clear definitions and structure in natural language), semi-formalization (introduction of symbols and formulas), and strong formalization (fully mathematized, computable). Mixed formalization is the normal state of theoretical evolution; mathematics is a sufficient but not necessary condition for science.

G

Gödel’s Incompleteness Theorems (*Ged’er buwanbei dingli*)

Proved by Kurt Gödel in 1931: any sufficiently powerful formal system is either incomplete or inconsistent. In this paper, these theorems are reinterpreted not as revealing a defect in mathematics but as the “license to exist” for any complex self-referential system—incompleteness prevents the system from falling into a logical black hole.

L

Logical Legitimacy

One of the four principles of legitimacy. A scientific theory must be internally consistent and free from contradiction. Derived from *System Ontology* Meta-Axiom 1 (the principle of systemic constraint), it is the minimal condition for any system to exist stably.

M

Meta-Axiom (*yuan gongli*)

The most fundamental presuppositions of *System Ontology*: Meta-Axiom 0 (the principle of self-

organization) and Meta-Axiom 1 (the principle of systemic constraint). Their coupling constitutes Meta-Theorem 1 (the eternal dynamical cycle of being). The four principles of this paper are derived from these meta-axioms and the system axioms.

Meta-Science (*yuan kexue*)

The theory of how science becomes science. This paper argues that such a theory must itself satisfy the four principles and, by adopting an external judgment stance, can achieve self-referential closure without vicious circularity, thereby qualifying as the highest form of science.

Meta-Theory (*yuan lilun*)

A theory about theories. This paper is a meta-theory about how philosophy becomes science, and it completes its own self-referential closure by passing the four-principle examination.

N

Noetic Ecology (*zhi zhi shengtaixue*)

One of the author's three works, the middle layer of the theoretical system. It investigates the laws of noetic systems at the cognitive, cultural, and social levels. The “cognitive expansion theorem” and the formalization of meta-cognitive power $M = T \times C \times R$ in that work provide the micro-level basis for the four principles in this paper.

O

Observability Legitimacy

One of the four principles of legitimacy. A scientific proposition must maintain a principled codifiable interaction channel with the cognitive framework (or designable measuring instruments), leaving detectable effects. Derived from *System Ontology* Axiom 3 (existence as interaction). It is distinguished from ontological “existence”—a system may exist (interacting with other systems) but still not satisfy observability legitimacy if it has no effective interaction channel with human cognition.

P

Paradigm (*fan shi*)

Kuhn's concept: the theoretical framework, fundamental assumptions, research methods, and exemplary achievements shared by a scientific community. In this paper, paradigms are subsumed under “cognitive frameworks,” and paradigm shifts are interpreted as cognitive escapes driven by the four-principle screening.

Potentiality Field (*qianneng yu*)

Denoted Φ . Defined as the set of all logically possible system laws, guaranteed by Meta-Axiom 0. Philosophy is the exploration of this field, unfolding the space of possibilities for science to screen.

R

Research Programme (*yanjiu gangling*)

Lakatos's concept: a scientific theory consists of an inviolable hard core and an adjustable protective belt. In this paper, research programmes are interpreted as cognitive frameworks; the hard core corresponds to the axiomatic layer, the protective belt to auxiliary hypotheses and formal expressions.

S

Self-Reference (*zi zhi*)

The capacity of a system to refer to or operate on itself. In formal systems, self-reference often leads to paradox; in hierarchically structured cognitive systems, it can be safely used at the meta-level. This paper argues that the legitimacy of self-reference depends on hierarchical difference, and the self-referential closure of the meta-theory is achieved through external judgment rather than internal self-proof.

Self-Referential Closure (*zi zhi bi huan*)

The state in which a meta-theory completes its own self-examination and establishes its legitimacy. The closure in this paper is benign: the meta-theory does not prove itself but is examined externally by the reader/author at the meta-level applying the four principles.

Semi-Formalization (*ban xingshihua*)

The middle level of formalization: specialized symbols and formulas are introduced, but the overall theory still relies partly on natural language. Core mechanisms are symbolized, but not all reasoning is fully formalized. The present theory is at this level.

Strong Formalization (*qiang xingshihua*)

The highest level of formalization: the theory is expressed entirely in mathematical language; all concepts have strict definitions; all reasoning can be formalized as computation or proof. Newtonian mechanics is a paradigm case.

Structural Tension (*jiegouxing zhangli*)

A concept from *System Ontology* Theorem 4: the inherent conflict between different levels of a system (e.g., subsystem interests vs. system-level stability) that drives evolution. In this paper, it explains why paradigms must eventually be replaced.

System Ontology (*xitong bentilun*)

One of the author's three works, the foundational layer of the theoretical system. It provides the ultimate principles of being (Meta-Axioms 0 and 1, six system axioms, seven core theorems) from which the four principles of this paper are derived.

T

Truth Subset (*zhenzhi ziji*)

Denoted Σ . Defined as $\Sigma = \{x \in \Phi \mid \text{Valid}(x)\}$, where $\text{Valid}(x)$ means the proposition or system law passes the four principles of legitimacy. Science is the systematic exploration of Σ .

W

Weak Formalization (*ruo xingshihua*)

The lowest level of formalization: natural language with clear definitions, explicit structure, and discernible rules. Darwinian evolution is a paradigm case, demonstrating that formalization is not equivalent to mathematization.